

Practical 25

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester II)
Scala for Data Science Practical

Practical - 25

Roll No. : S006	Name:Seja Manohar Deshmukh
Class :SYIT	Batch:01
Date of Assignment :11/08/2026	Date/Time of Submission :11/08/2026

Practical 25

Aim: Handle Missing values in a dataset. Replace missing values with column mean.

The screenshot shows an IDE with a project named 'Prac26'. The code in 'Prac26.scala' defines an object 'Prac26' extending 'App'. It creates a list 'sales' with values [100.0, 200.0, null, 300.0, null]. It calculates the mean of non-null values (200.0) and replaces the null values in the 'sales' list with this mean. The output in the Run console shows the mean sales, the original data with nulls, and the data after replacement.

```
object Prac26 extends App {  
  1  
  2  
  3  val sales: List[java.lang.Double] =  
    List(100.0, 200.0, null, 300.0, null)  
  4  
  5  
  6  val v = sales.filter(_ != null)  
  7  
  8  val mean = v.map(_.toDouble).sum / v.length  
  9  
 10  println("Mean Sales: " + mean)  
 11  
 12  val p = sales.map { value =>  
    13    if (value == null) mean  
    14    else value  
    15  }  
 16  
 17  println("Original Data:")  
 18  println(sales.mkString(", "))  
 19  
 20  println("After Replacing Missing Values:")  
 21  println(p.mkString(", "))  
 22  println("S006 Sejal Deshmukh")  
 23}
```

Run console output:

```
Mean Sales: 200.0  
Original Data:  
100.0, 200.0, null, 300.0, null  
After Replacing Missing Values:  
100.0, 200.0, 200.0, 300.0, 200.0  
S006 Sejal Deshmukh
```

Practical 26

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Scala for Data Science Practical

Practical - 26

Roll No. : S006	Name:Seja Manohar Deshmukh
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Practical 26

Aim: Perform logistic regression using Breeze. Classify a dataset with binary labels.

```
build.sbt  Prac26.scala x
1  import breeze.linalg._
2  ▶ object Prac26 extends App {
3
4      val x = DenseVector(1.0, 2.0, 3.0, 4.0, 5.0) // Dataset
5
6      val y = DenseVector(0.0, 0.0, 0.0, 1.0, 1.0) // Binary labels
7
8      var weight = 0.0
9      var bias = 0.0
10     // Sigmoid function
11     def sigmoid(z: Double): Double = {
12         1.0 / (1.0 + math.exp(-z))
13     }
14     for (i <- 1 to 1000) {
15         for (j <- 0 until x.length) {
16             val prediction = sigmoid(weight * x(j) + bias)
17             val error = prediction - y(j)
18             weight = weight - 0.01 * error * x(j)
19             bias = bias - 0.01 * error
20         }
21     }
22     // Classify the data
23     println("Predictions:")
24     for (i <- 0 until x.length) {
25         val probability = sigmoid(weight * x(i) + bias)
26         val result =
27             if (probability >= 0.5) 1 else 0
28         println("Input = " + x(i) + ", Probability = " + probability + ", Class = " + result)
29     }
30     println("S006 Sejal Deshmukh")
31 }
```

```
"C:\Program Files\Java\jdk1.8.0_341\bin\java.exe" ...
```

```
Predictions:
```

```
Input = 1.0, Probability = 0.0515501057576029, Class = 0
```

```
Input = 2.0, Probability = 0.16412826954644827, Class = 0
```

```
Input = 3.0, Probability = 0.4149888799701166, Class = 0
```

```
Input = 4.0, Probability = 0.7193154698480789, Class = 1
```

```
Input = 5.0, Probability = 0.9025176083649726, Class = 1
```

```
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```

```
Process finished with exit code 0
```

```
|
```